EPA TADA for Efficient and Reproducible Water Quality Analyses: Project Overview and R Workflow Demo Using Penobscot Nation's Publicly Available WQP Data

Cristina Mullin

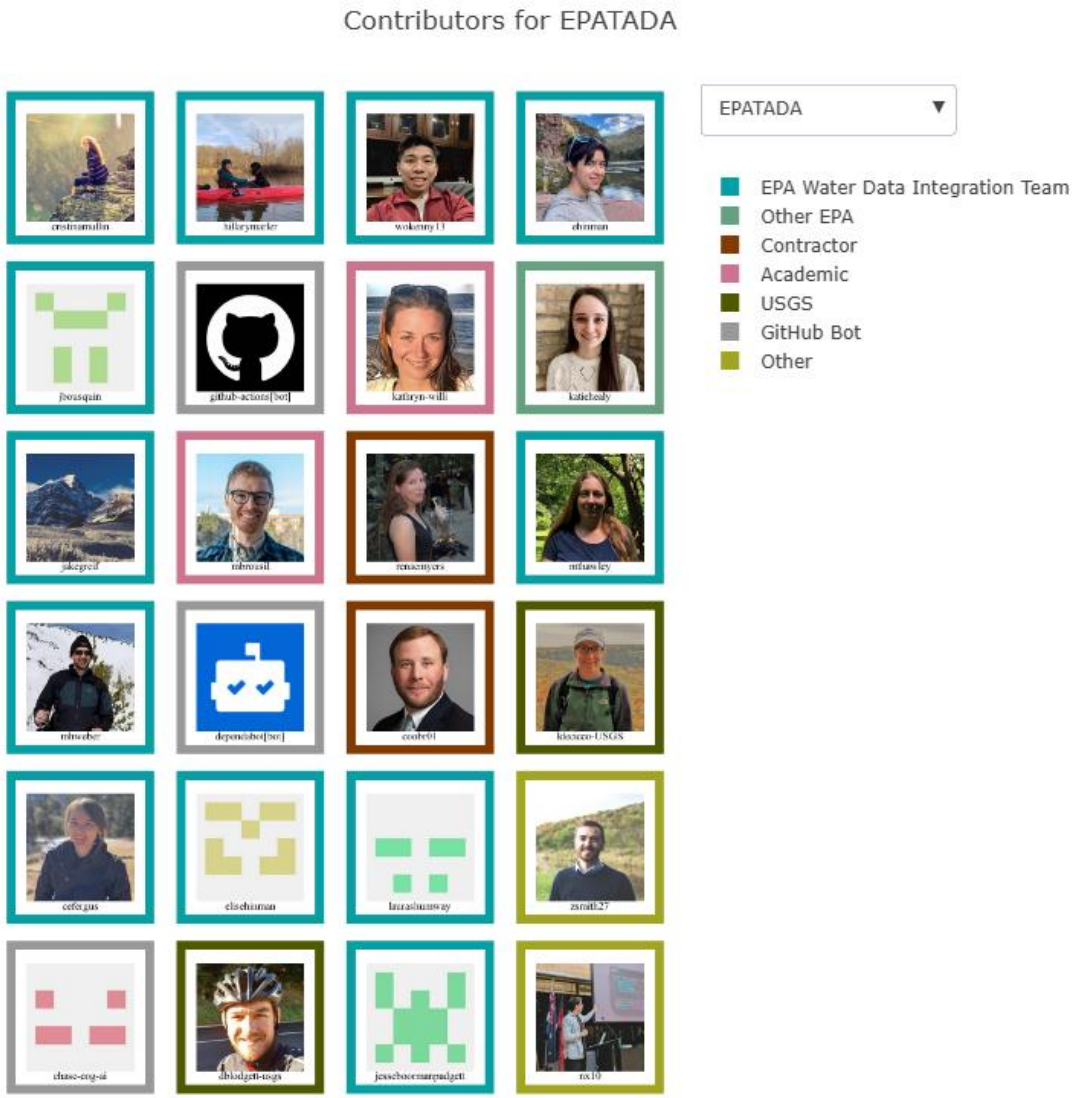
March 30, 2026

Agenda

- Introductions – EPA Water Data Integration Team
- TADA Overview
- TADA Module 1: WQX/WQP Data Discovery and Cleaning
- TADA Module 2: ATTAINS Assessment Unit and Use Integration
- TADA Module 3: Define Water Quality Criteria and Analyze
- Questions

TADA Team

- EPA TADA Team
  - Cristina Mullin
  - Hillary Marler
  - Kenny Wong (ORISE Participant)
  - Justin Bousquin
  - Emi Fergus
  - Marc Weber
  - Jesse Boorman-Padgett
  - Justin Drew
  - Sarah Wheeler (R8)
  - Kevin Christian



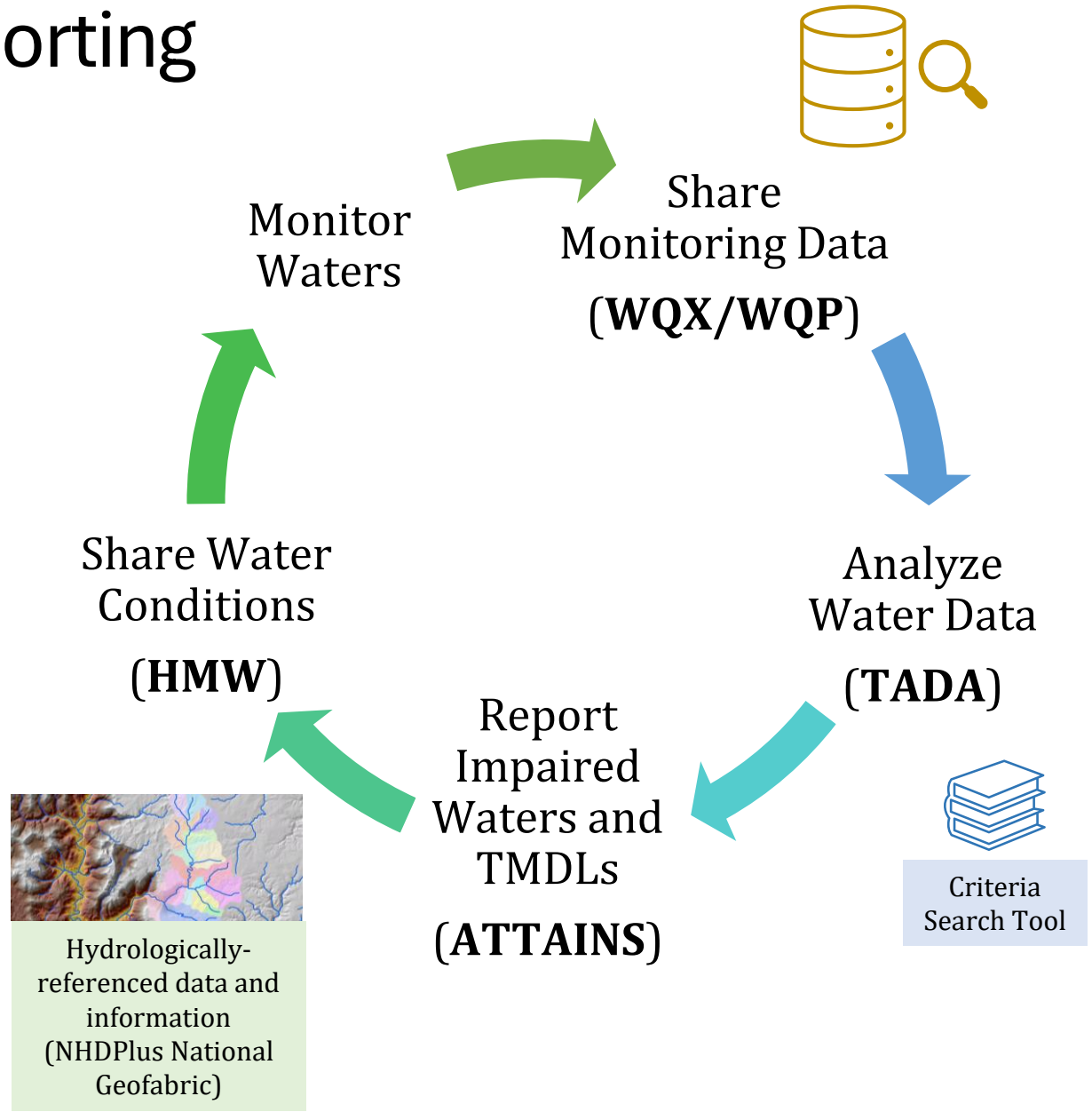
Data Systems and Tools Supporting the Clean Water Act (CWA)

Water Data integration Team Systems & Tools

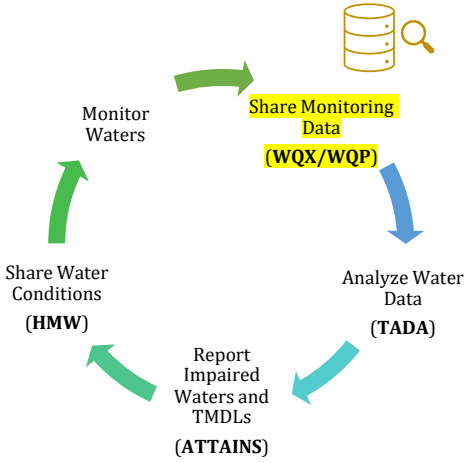
- Water Quality eXchange (WQX)
- Water Quality Portal (WQP)
- Tools for Automated Data Analysis (TADA)
- Assessment and Total Maximum Daily Load Tracking and Implementation System (ATTAINS)
- How's My Waterway? (HMW)
- Grants Reporting and Tracking System (GRTS)
- Stream-Catchment (StreamCat) Dataset
- National Hydrography Dataset Plus (NHDPlus)

Statutory and Administrator Priorities

- CWA: 106(e), 319, 305(b), 303(d), 604(b)
- Pillars: 1, 3 and 4



WQX – To Share Monitoring Data

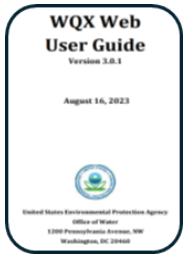


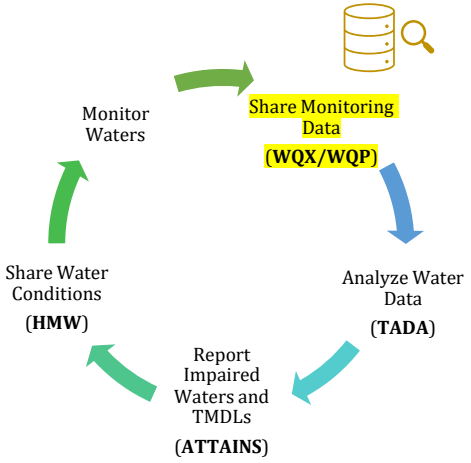
- Launched in 2009 to replace STORET and serves as the primary framework for organizations to share water monitoring data with EPA
- Includes water monitoring data as recent as this week and going back as far as the 1800'
- Provides EPA partners a way to fulfill statutory data reporting requirements when using CWA funds
- More than a database! – WQX is a common data model following universal data standards to share diverse monitoring data across different organizations
- Includes biological (taxon), chemical, and physical data & multiple media types – water, air, sediment, biological tissue
- Various tools available to streamline data sharing with minimal data management demands on provider

WQX Resources

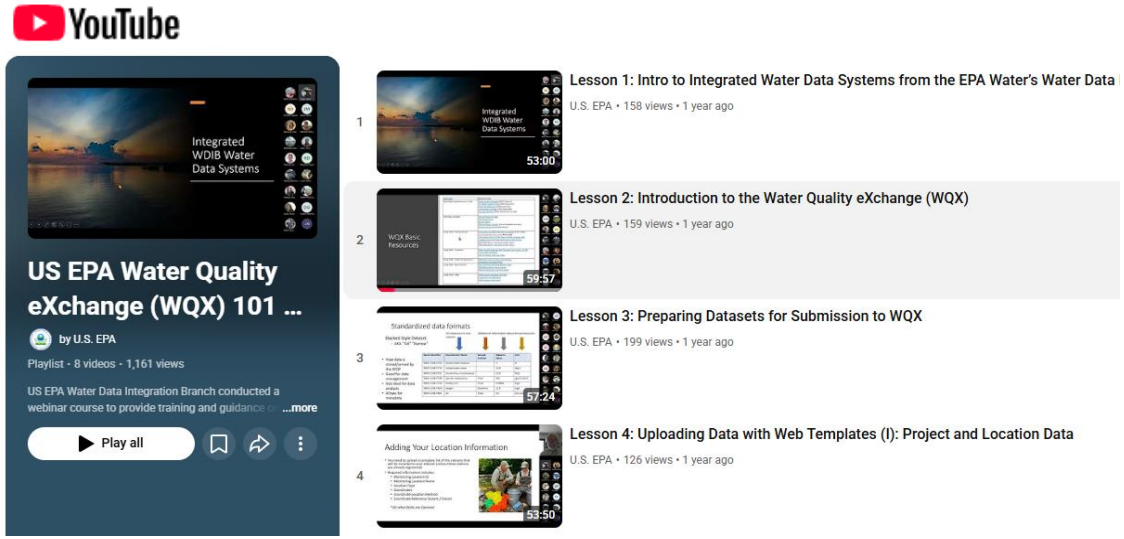
- EPA WQX Website
  - Description of WQX
  - Getting started
  - Links to WQX schema & domain values

- WQX Web User Guide

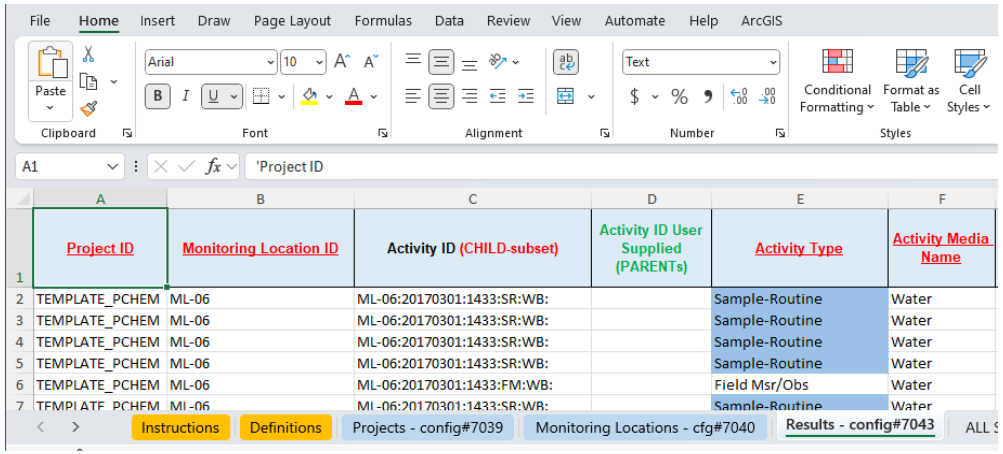




- WQX 101 YouTube Videos

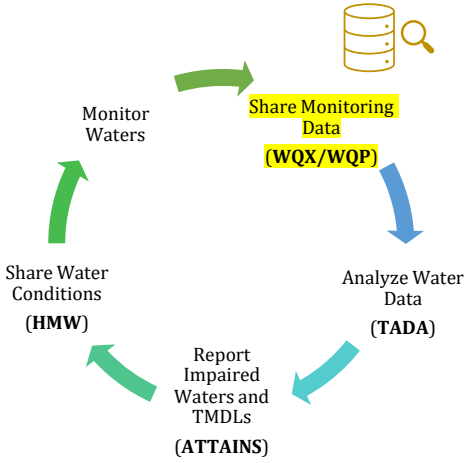


- WQX Web Excel Templates

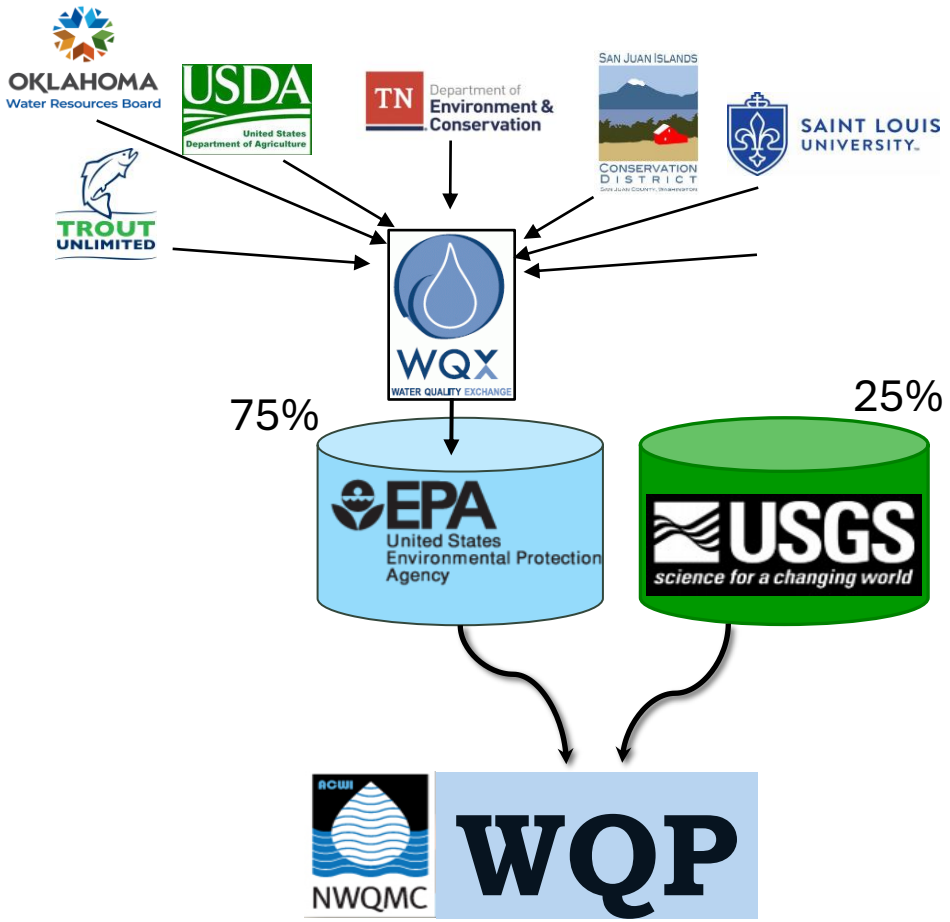


	Project ID	Monitoring Location ID	Activity ID (CHILD-subset)	Activity ID User Supplied (PARENTs)	Activity Type	Activity Media Name
2	TEMPLATE_PCHEM	ML-06	ML-06:20170301:1433:SR:WB:		Sample-Routine	Water
3	TEMPLATE_PCHEM	ML-06	ML-06:20170301:1433:SR:WB:		Sample-Routine	Water
4	TEMPLATE_PCHEM	ML-06	ML-06:20170301:1433:SR:WB:		Sample-Routine	Water
5	TEMPLATE_PCHEM	ML-06	ML-06:20170301:1433:SR:WB:		Sample-Routine	Water
6	TEMPLATE_PCHEM	ML-06	ML-06:20170301:1433:FM:WB:		Field Msr/Obs	Water
7	TEMPLATE_PCHEM	ML-06	ML-06:20170301:1433:SR:WB:		Sample-Routine	Water

WQP - To Access Water Quality Data

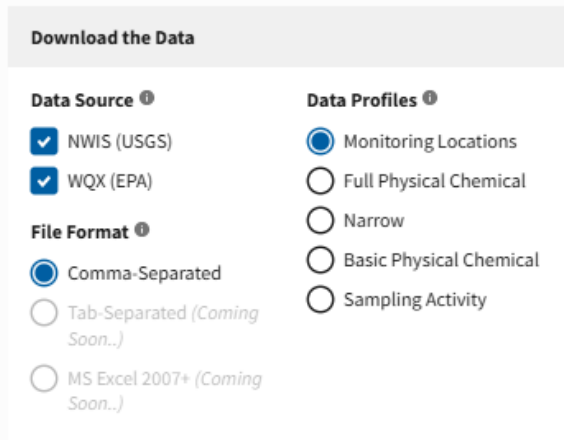


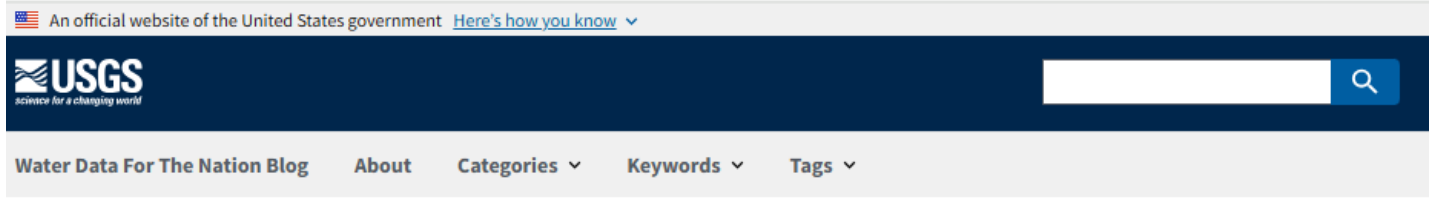
- **Vision:** Premiere source for water data for everyone, everywhere.
- **Usage:** 440M+ records from 1.1M+ sites by 1400+ organizations. Most used water quality dataset in the US (4200 unique users per day, billions of rows of data served to users each month, ~250% increase year-over-year). Diverse data contributors: federal agencies (USGS, EPA, USDA), state agencies, tribes, NGOs, universities, participatory scientists.
- **Formats:** Combines cross-organizational data in standard format (operates under Interagency Agreement with EPA and USGS). Available for download or via web services.
- **Interface & Search:** Easy search, filter, and retrieval by location, date, and parameters.
- **Value:** Allows contributors to cut their own database costs while improving data quality and availability nationally. Allows for data reuse beyond the original project. Available for analysis by CWA community, regulatory users (e.g., permit writers), researchers, policymakers, and the public.
- **Tools:** HMW and R packages (dataRetrieval, EPATADA) for efficient retrieval, quality control, and analysis.



WQX/WQP 3.0 Beta

- Try WQP Beta & Provide Feedback!
  - Water Quality Portal Quick Reference Guide
  - WQX3.0 Data Now Available on the Water Quality Portal
  - EPATADA R Package Guidance on WQX 3.0 Migration





WQX3.0 Data Now Available on the Water Quality Portal

WQX3.0 data are available through the Water Quality Portal. This updated version of the schema includes new elements and new profiles with updated column names.

DATE POSTED
May 20, 2024

LAST UPDATED
November 25, 2025

AUTHOR
Candice Hopkins
Lee Stanish
Cristina Mullin (USEPA)
Adam Griggs (USEPA)

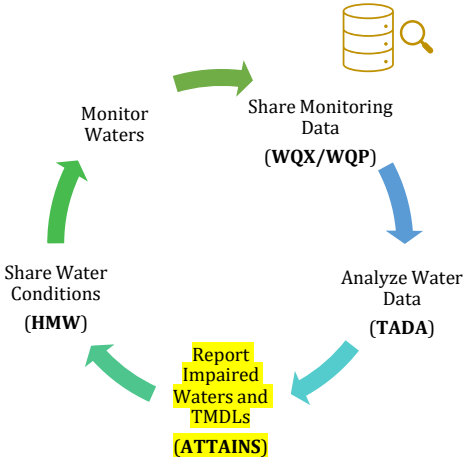
READING TIME
5 minutes

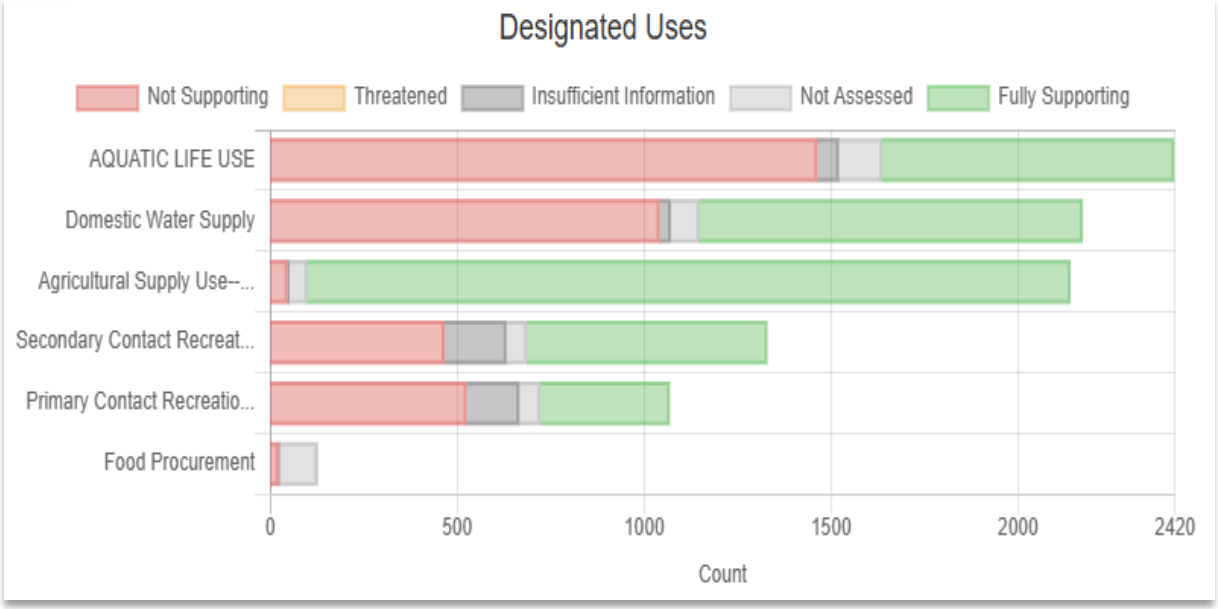
SHARE

What's on this page

How is this different from what you delivered before?
Where will I see these new data elements?
How do I update my workflow from WQX2.2 to WQX3.0?
Will the previous WQX2.2 data profiles still be available?
I have a current tool I've built on WQP datasets. Will I need update my workflow from 2.2 to 3.0.?
Community Development of WQX3.0
Feedback

Assessment TMDL Tracking and Implementation System (ATTAINS)

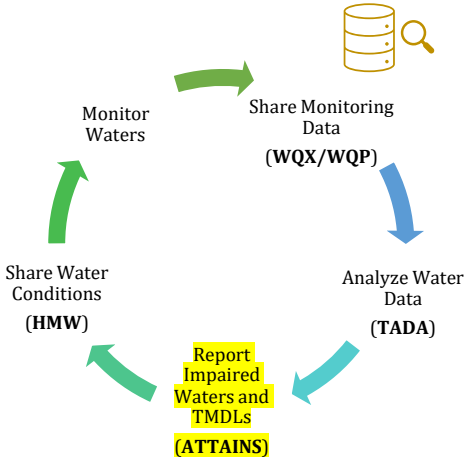




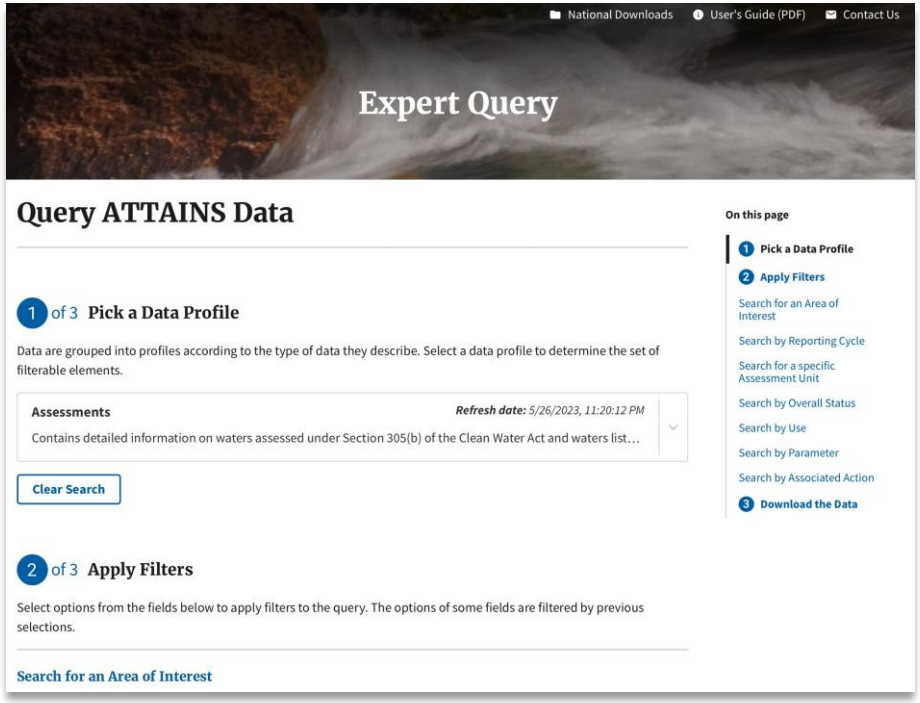
An ATTAINS Report for Kansas 2024 Reporting Cycle

- Waterbodies & Uses - Assessment Unit names, water types, sizes, geospatial locations
- Assessment decisions – Integrated Reporting (IR) = CWA Sections 303(d) and 305(b)
- Actions or Plans towards restoration or protection of waters (e.g., TMDLs, Advance Restoration Plans (ARPs), Protection Approaches)
- State statistical survey results
- Submitters include EPA, states, and participating tribes

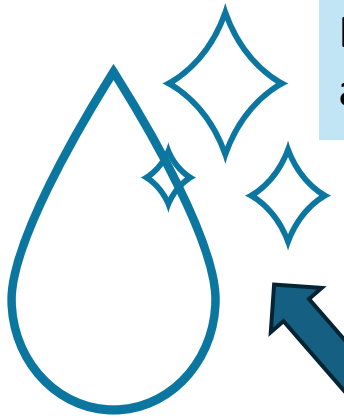
ATTAINS Data Retrieval



- Geospatial data
  - Most downloaded service on EPA's GIS server
  - Gets over 80% of traffic out of 276 services
- Public Web Services
  - Populates How's My Waterway, state applications
- Expert Query
  - User interface for accessing ATTAINS data in tabular format.
  - Advanced cross-organization and national-level services and query options.
  - Allows Actions document text search.
- **rExpertQuery R Package**

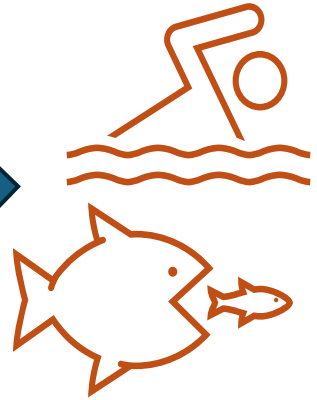


Basics of Water Quality Assessment

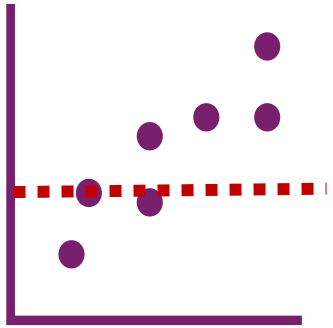


Discover, assemble, and QC available data

Assign assessment units and uses



Assessment unit overlay with monitoring locations



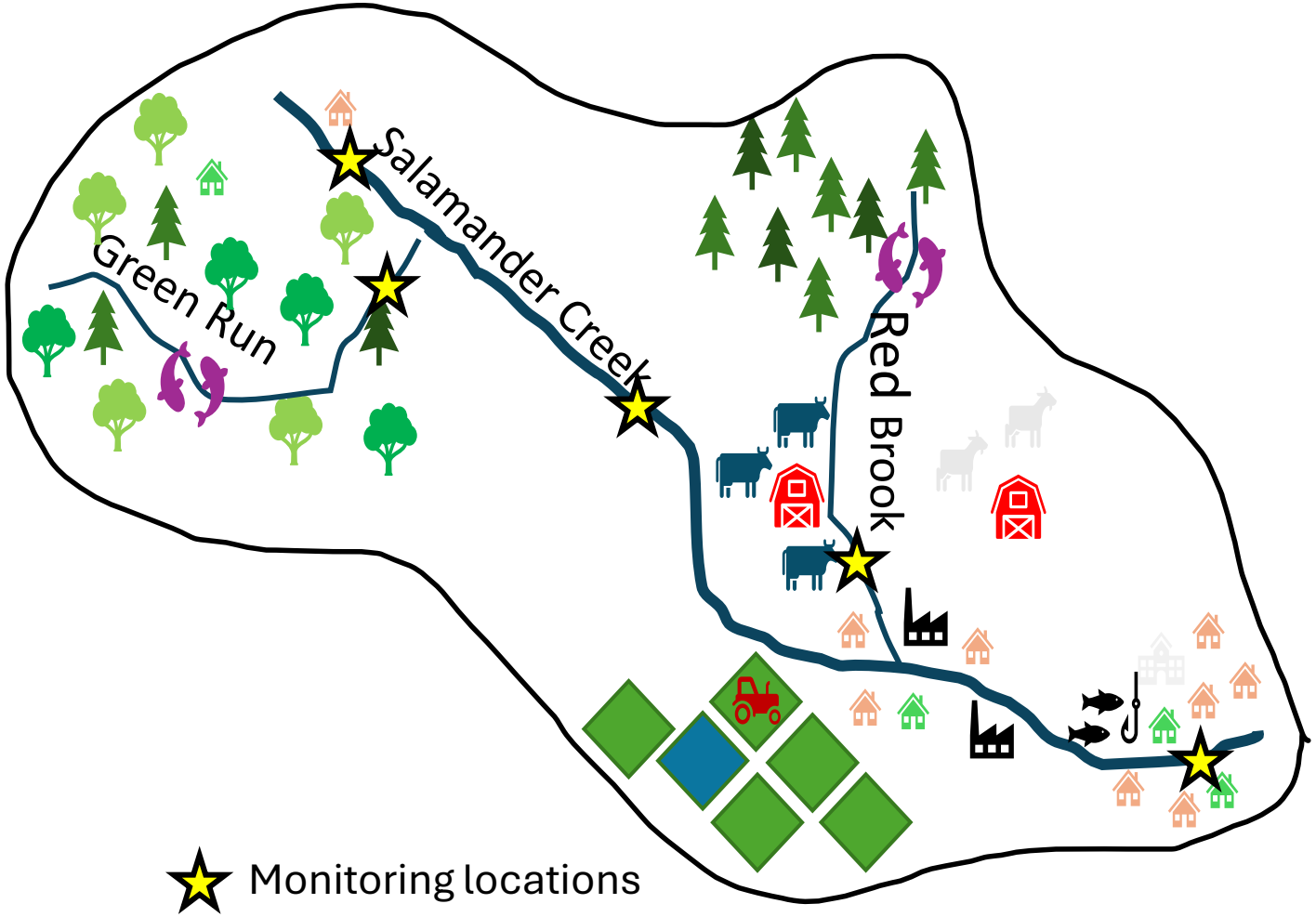
Designated uses drive numeric and narrative water quality criteria

Data analysis guides impairment decisions based on:

- Criteria components
  - Frequency
  - Duration
  - Magnitude
- Additional complexities
  - Seasonality
  - Site-specific
  - Acute vs. chronic exposure
  - Spatial & temporal aggregation
  - Sampling depth
  - And more!

Example: Salamander Creek Watershed

Type	Designated Use	Salamander Creek	Red Brook	Green Run
Aquatic Life Support	Aquatic Life Other Than Fish	X	X	X
	Coldwater Fisheries		X	X
	Warmwater Fisheries	X		
Recreation	Recreation		X	X
Water Supply	Irrigation	X		
	Livestock		X	
	Potable water	X		



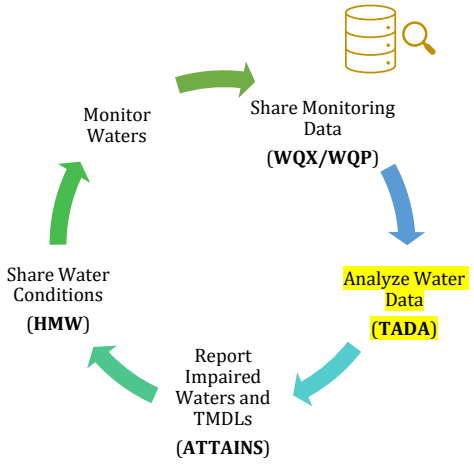
Salamander Creek: Numeric Criteria

Salamander Creek is designated for four uses that have the following numeric criteria:

Parameter	Unit	Type	Statistic	Exceedance	Aquatic Life Other Than Fish	Irrigation Water Supply	Public Drinking Water Source	Warmwater Habitat
Conductivity	uS/cm	Maximum	Instantaneous	10%	750	2,500	1,000	1,500
Dissolved oxygen	mg/L	Minimum	Instantaneous	None	5.0	--	--	5.0
Nitrate	mg/L	Maximum	Average*	None	1.5	100	10*	1.0
pH	SU	Range	Instantaneous	None	6.5<pH<9	--	--	6.5<pH<9
Total phosphorus	mg/L	Maximum	Average	None	0.1	--	--	0.3

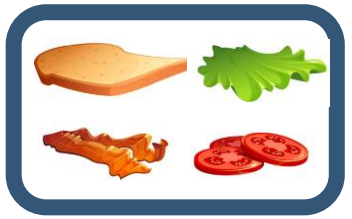
** The nitrate criterion is instantaneous for the public drinking water source.*

TADA: Tools for Automated Data Analysis

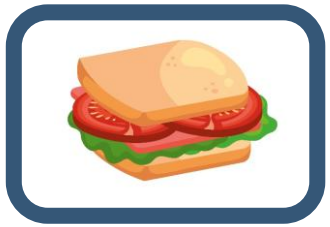


- A suite of open-source R tools designed to efficiently discover, compile, clean, analyze, and visualize data from the WQP and ATTAINS
- A hub for the open-source water quality community to collaborate!
- Reproducible workflows
- R Shiny Learning Resources

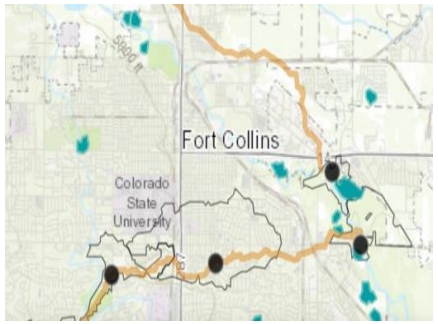
- Flexible Design



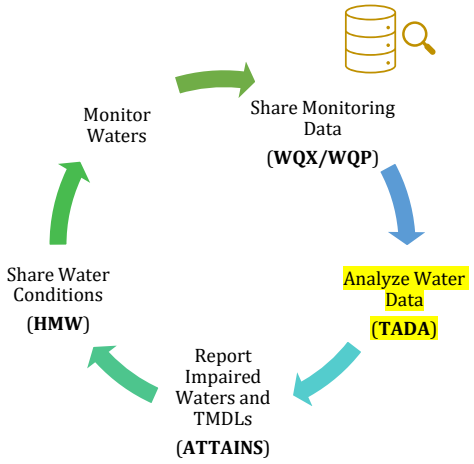
TADA Functions



Your Analysis

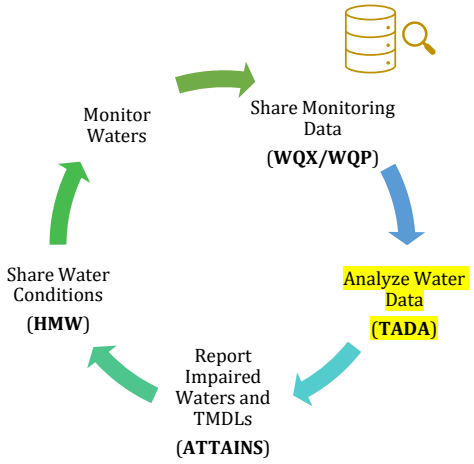


TADA Limitations and Challenges



- Does not cover narrative/text data
- Water data only for now (could expand in future)
- Does not include WQP biological data profile (taxon)
- Will require user upkeep / maintenance
- User decides which components to use (or not)
- Discrete data only for now (could expand in future)
- Human review required
- Cannot fully standardize or automate the process (users must supply details & make individual decisions along the way)

Broader Impacts



Use of TADA has potential to greatly reduce total government costs across

- State and tribal agencies
- EPA regions & HQ
- USGS, other federal agencies
- National water quality community

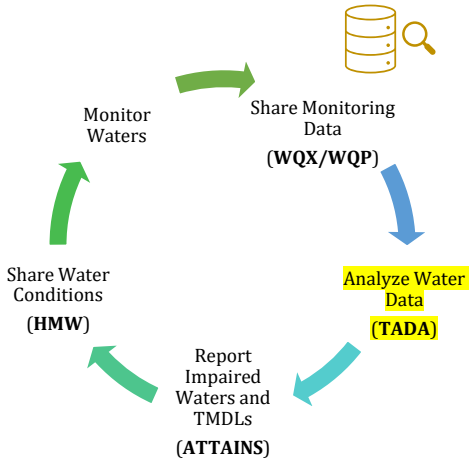
Efficient, transparent, and reproducible assessments

- Frees up time for other important tasks
- May facilitate assessing more waters
- Helps discover and share commonalities in assessment processes nationally

Building data equity

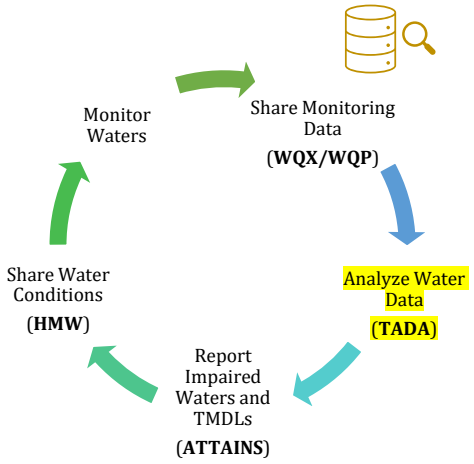
- Facilitates use of other organizations data in State or Tribal assessments
- Makes the WQX QAQC service available on the WQP side
- Helps find and address data quality issues in WQX/WQP

TADA Working Group Goals



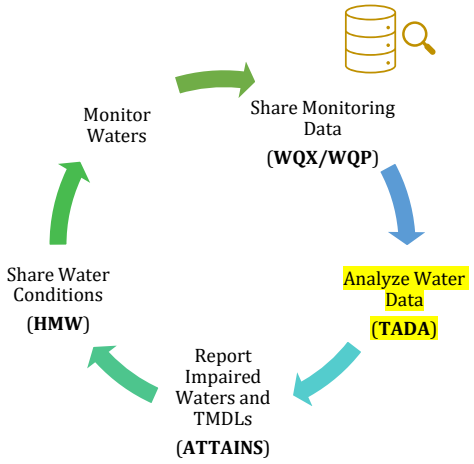
- Build and share reproducible, open-source R code to integrate, evaluate, and visualize data from the Water Quality Portal (WQP), the Assessment, Total Maximum Daily Load (TMDL) Tracking and Implementation System (ATTAINS), and the Criteria Search Tool.
- Improve the efficiency, transparency, and reproducibility of analyses.
- Create flexible, modular tools that integrate with existing workflows and are easy to customize.
- Learn R together and ensure products are usable and accessible—especially for under-resourced organizations.

How to participate



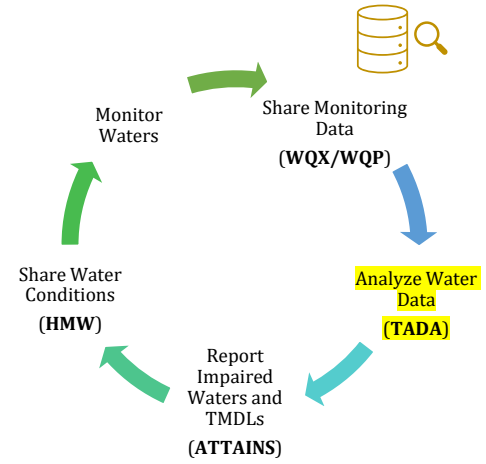
- Pre-submit questions: reply in the Teams meeting chat or email us.
- Want to showcase your work? Volunteer or nominate a User Spotlight.
- Join via Microsoft Teams
- Notes/slides (and recordings, when available) will be shared within 48 hours.
- All skill levels welcome; familiarity with R is helpful but not required.
- Contact: mywaterway@epa.gov or mullin.cristina@epa.gov

Schedule



- Cadence: Every 6 weeks on Tuesdays (e.g., Mar 10, Apr 21, Jun 2, 2026)
- Time: 2–3 PM ET
- Demo + Q&A (focused feature/workflow or a User Spotlight)
- Office Hours (pre-submitted questions first, then open Q&A)

You can help develop TADA!



TADA 0.0.1 Reference Articles Search for

Contributing

2023-06-07

Source: vignettes/CONTRIBUTING.Rmd

Contribute to TADA!

We encourage you to read this project's CONTRIBUTING policy (you are here), its LICENSE, and its README.

We're so glad you're thinking about contributing to an EPA open source project! If you're unsure about anything, just ask — or submit your issue or pull request anyway. The worst that can happen is we'll politely ask you to change something. We appreciate all friendly contributions.

No matter who you are, if you spot an error, omission, or bug, you're welcome to open an issue in this repo!

TADA Working Group Mission

To share and develop R code for evaluating and visualizing Water Quality Portal (WQP) data more efficiently though collaboration and open-source programming. This includes working together to find commonalities in assessment processes across the nation, creating flexible tools that can be easily customized to work within existing workflows, supporting each other in learning R, and ensuring products will be accessible to organizations most in need.

On this page

- Contribute to TADA!
- TADA Working Group Mission
- Package Development
- What is GitHub?
- Required Installations
- Issues
- Branches and Pull Requests
- Additional References
- Open-Source Code Policy
- License
- Disclaimer
- Contact

Review required
At least 1 approving review is required by reviewers with write access. Learn more

Some checks haven't completed yet
5 successful and 1 in progress checks

- ✓ R-CMD-check / macos-latest (release) (pull_request) Successful in 52m
- ✓ test-coverage / test-coverage (pull_request) Successful in 10m
- ✓ R-CMD-check / windows-latest (release) (pull_request) Successful in 43m
- R-CMD-check / ubuntu-latest (devel) (pull_request) *In progress — This check*
- ✓ R-CMD-check / ubuntu-latest (release) (pull_request) Successful in 35m
- ✓ R-CMD-check / ubuntu-latest (oldrel-1) (pull_request) Successful in 25m

Contributors 24

+ 10 contributors

Protect matching branches

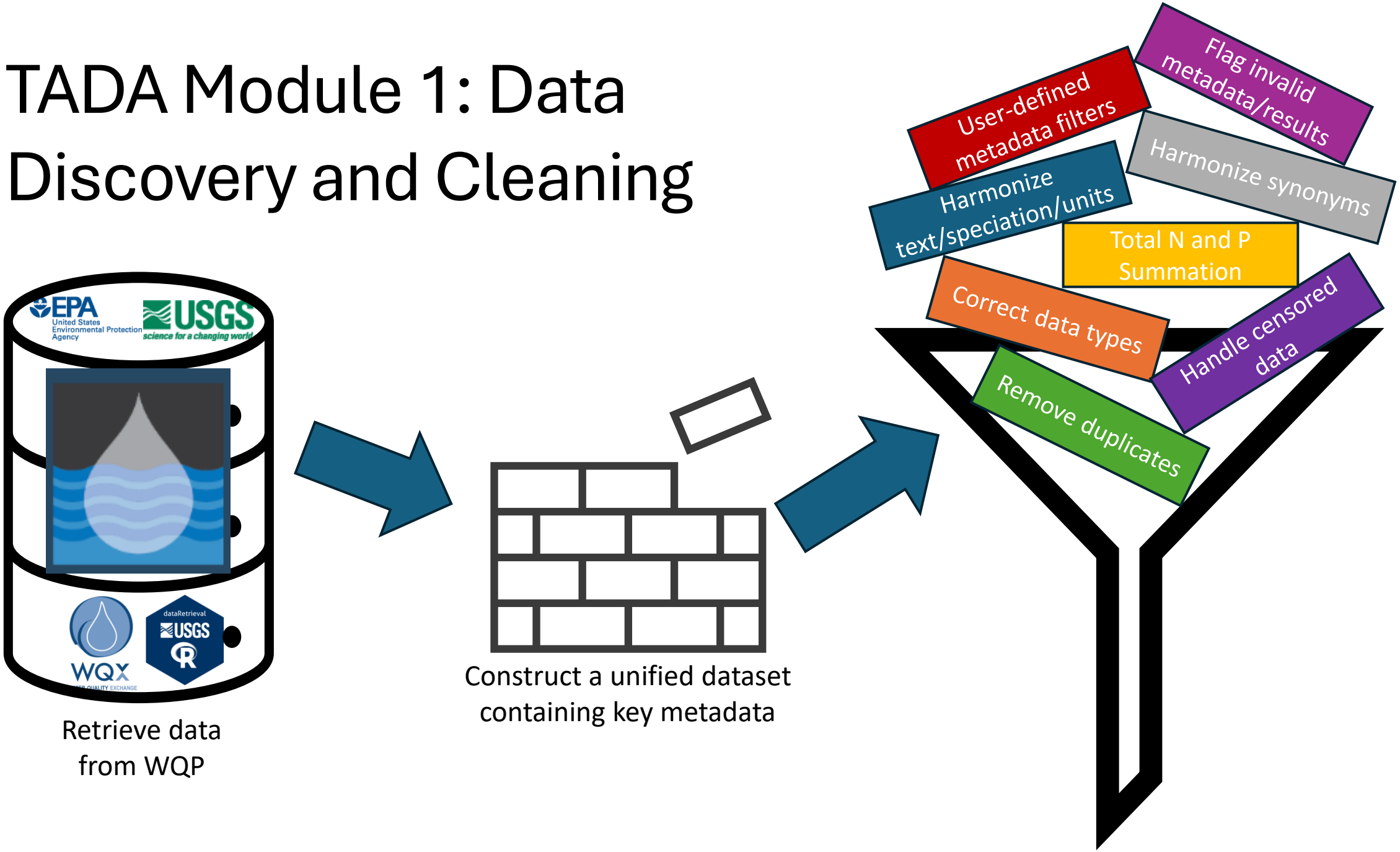
☑ **Require a pull request before merging**
When enabled, all commits must be made to a non-protected branch and submitted via a pull request before they can be merged into a branch that matches this rule.

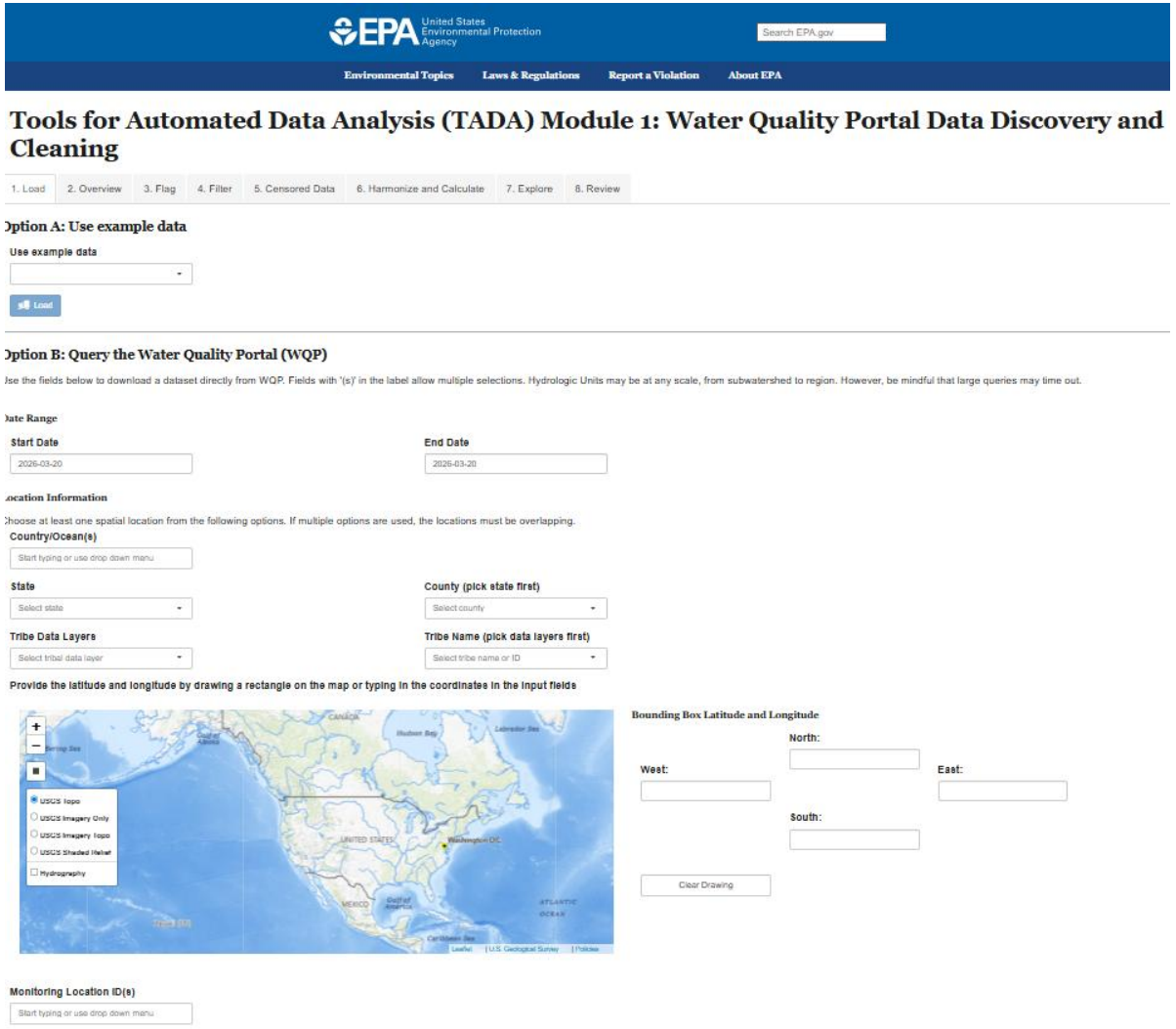
☑ **Require approvals**
When enabled, pull requests targeting a matching branch require a number of approvals and no changes requested before they can be merged.

Required number of approvals before merging: 1

Status of TADA Modules 1-3

TADA Module 1: Data Discovery and Cleaning





https://rconnect-public.epa.gov/TADAShiny/

EPATADA 0.0.1

TADA Module 1: Training for Beginner R Users

TADA Team

2026-03-02

Source: `vignettes/TADAModule1_BeginnerTraining.Rmd`

R5 Profile

The table above shows the first 100 records from the R5 Water Quality Portal Query.

Question 1: How many results did TADA_DataRetrieval return?

```
# Determine number of rows in the data set
R5_nresults <- nrow(R5Profile)
```

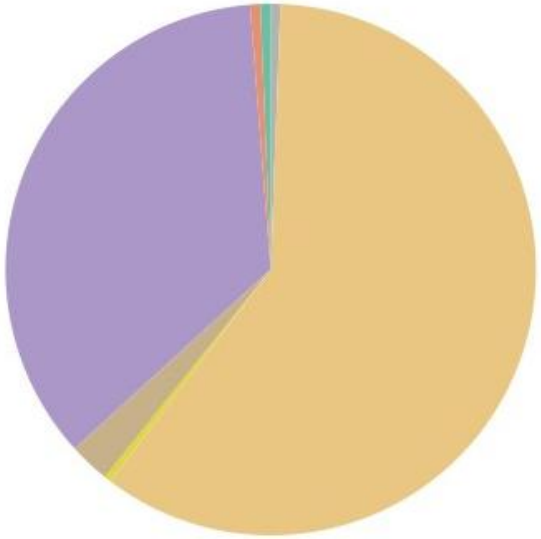
Review and Filter By Media Type

TADA is primarily designed to accommodate water data from the WQP. Let's see what activity media types are represented in the data set.

https://usepa.github.io/EPATADA/articles/TADAModule1_BeginnerTraining.html

Summary of User Decisions

TADA.Remove	TADA.RemovalReason
TRUE	Flag: Measurement activity type code indicates it is a QC replicate, duplicate, or blank
TRUE	Filter: Exclude ActivityTypeCode is Sample-Composite Without Parents
TRUE	Filter: Exclude HydrologicCondition is Rising stage or Falling stage
FALSE	
TRUE	Filter: Exclude MonitoringLocationTypeName is Wetland Undifferentiated or Well or Waste Sewer or Storm Sewer or Land or Facility Other or Canal Transport or Canal Irrigation or Canal Drainage
TRUE	Flag: Result value is not numeric or result value is NA and no detection limit value is provided, Flag: Measurement activity type code indicates it is a QC replicate, duplicate, or blank
FALSE	
TRUE	Flag: Result value is not numeric or result value is NA and no detection limit value is provided
TRUE	Flag: Measurement activity type code indicates it is a QC replicate, duplicate, or blank, Filter: Exclude MonitoringLocationTypeName is Wetland Undifferentiated or Well or Waste Sewer or Storm Sewer or Land or Facility Other or Canal Transport or Canal Irrigation or Canal Drainage



Removal Reasons

- Filter: Exclude ActivityTypeCode is Sample-Composite Without Parents - 18 results
- Filter: Exclude HydrologicCondition is Rising stage or Falling stage - 18 results
- Filter: Exclude MonitoringLocationTypeName is Wetland Undifferentiated or Well or Waste Sewer or Storm Sewer or Land or Facility Other or Canal Transport or Canal Irrigation or Canal Drainage - 1028 results
- Filter: Exclude TADA.CharacteristicName is AMMONIA - 72 results
- Flag: Conflict between detection condition text and detection limit type or detection limit type is not in WQX domain tables (likely USGS/NWIS-specific) - 12 results
- Flag: Measurement activity type code indicates it is a QC replicate, duplicate, or blank - 1729 results
- Flag: Result value is not numeric or result value is NA and no detection limit value is provided - 17 results

TADA Module 1: Data Discovery and Cleaning

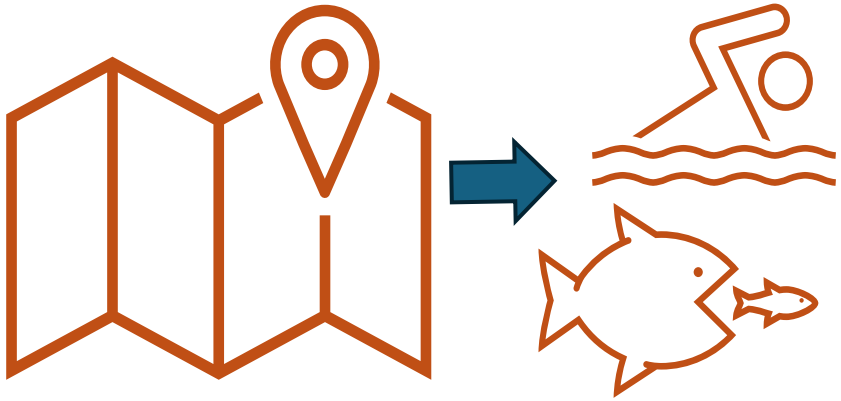
Three R Package Vignettes

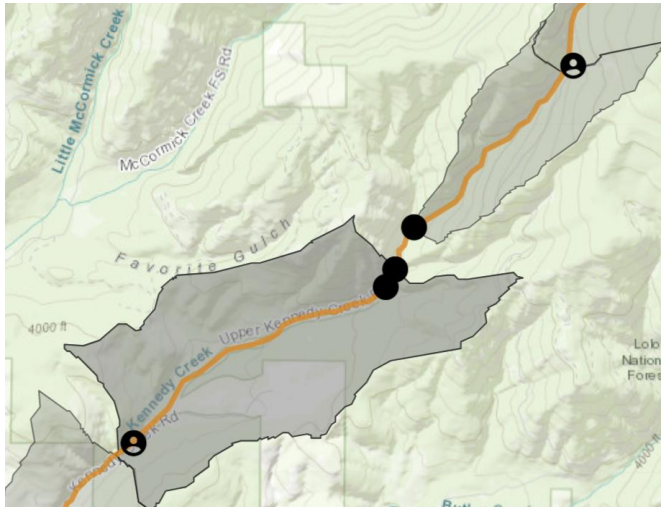
- Water Quality Portal Data Discovery and Cleaning
- Training for Beginner R Users
- Training for Intermediate/Advanced R Users

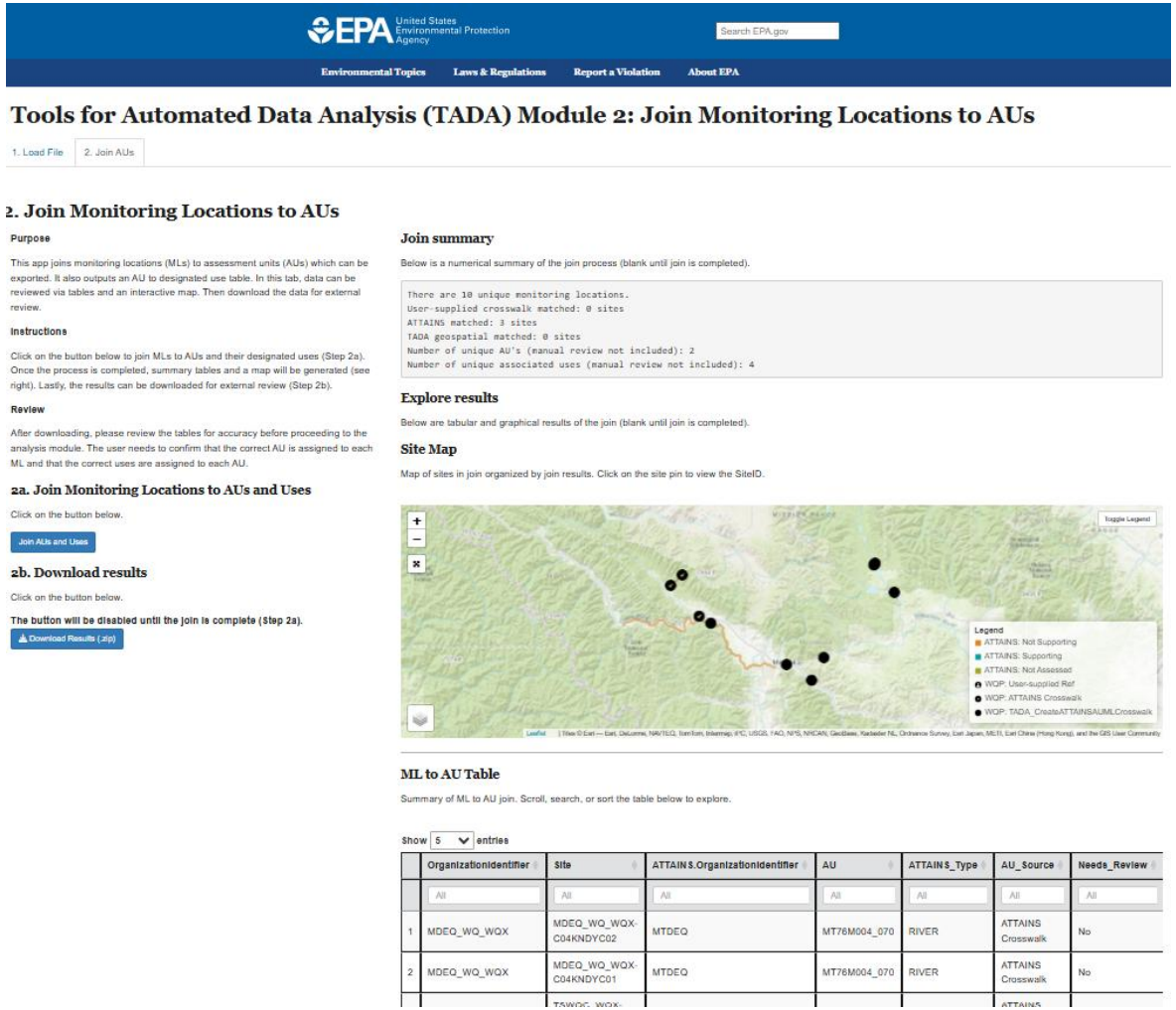
R Shiny App

- Public R Shiny App (only making minor updates, ready for use!)
- GitHub Repository

TADA Module 2: ATTAINS Assessment Unit and Use Integration







https://rconnect-public.epa.gov/TADAShinyJoinToAU/

EPATADA 0.0.1

TADA Module 2: Geospatial Functions

TADA Team

2026-03-02

Source: vignettes/TADAModule2.Rmd

Run `TADA_CreateAUMLCrosswalk`

```
# make AU assignments for unassigned MLs
MT.AUMLRef <- TADA_CreateAUMLCrosswalk(
  tada.MT.clean,
  au_ref = user.supplied.cw,
  org_id = "MTDEQ",
  fill_ATTAINS_catch = TRUE,
  return_nearest = TRUE,
  batch_upload = TRUE,
  api_key = api_key
)
```

```
# extract ATTAINS_crosswalk data frame from the list and show tab
TADA_TableExport(MT.AUMLRef$ATTAINS_crosswalk)
```

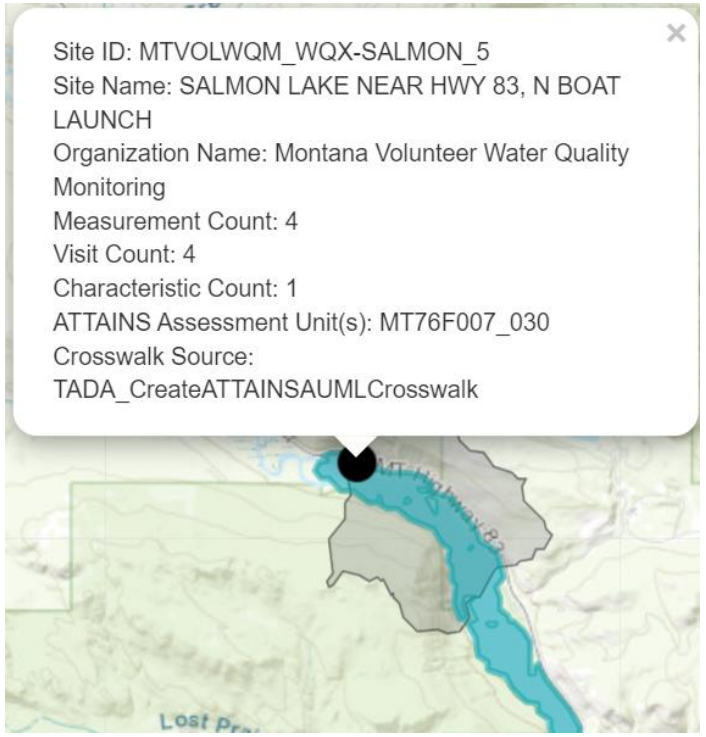
https://usepa.github.io/EPATADA/articles/TADAModule2.html

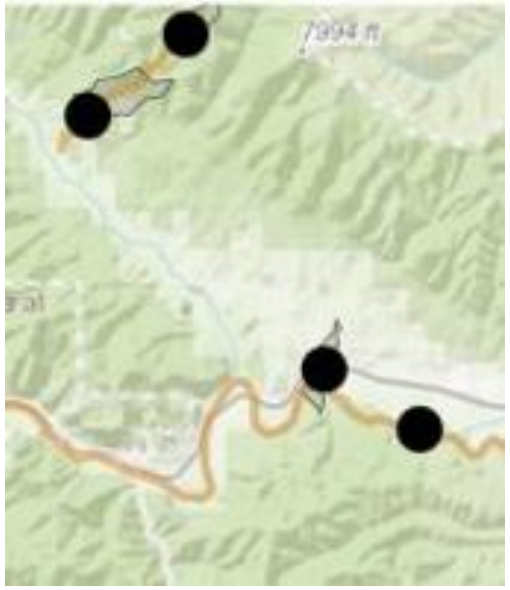
ML to AU Table

Summary of ML to AU join. Scroll, search, or sort the table below to explore.

Show 10 entries

	OrganizationIdentifier	Site	ATTAINS.OrganizationIdentifier	AU	ATTAINS_Type
	All	All	All	All	All
1	USGS-MT	USGS-12334550	MTDEQ	MT76E001_010	RIVER
2	USGS-MT	USGS-12340500	MTDEQ	MT76M001_030	RIVER
3	USGS-MT	USGS-12340000	MTDEQ	MT76F001_033	RIVER





AU to Use Table

Summary of AU to Use join. Scroll, search, or sort the table below to explore.

Show 5 entries

	ATTAINS.OrganizationIdentifier	ATTAINS.AssessmentUnitIdentifier	ATTAINS.UseName	ATTAINS.WaterType	TADA.AssessmentUnitStatus
	All	All	All	All	All
1	MTDEQ	MT76E001_010	Agricultural	RIVER	Existing
2	MTDEQ	MT76E001_010	Aquatic Life	RIVER	Existing
3	MTDEQ	MT76E001_010	Drinking Water	RIVER	Existing

TADA Module 2: ATTAINS Assessment Unit and Use Integration

One R Package Vignette

- Geospatial Functions

R Shiny App

- R Shiny Application (still under development, open to early feedback, ready for use by summer 2026):
- GitHub Repository

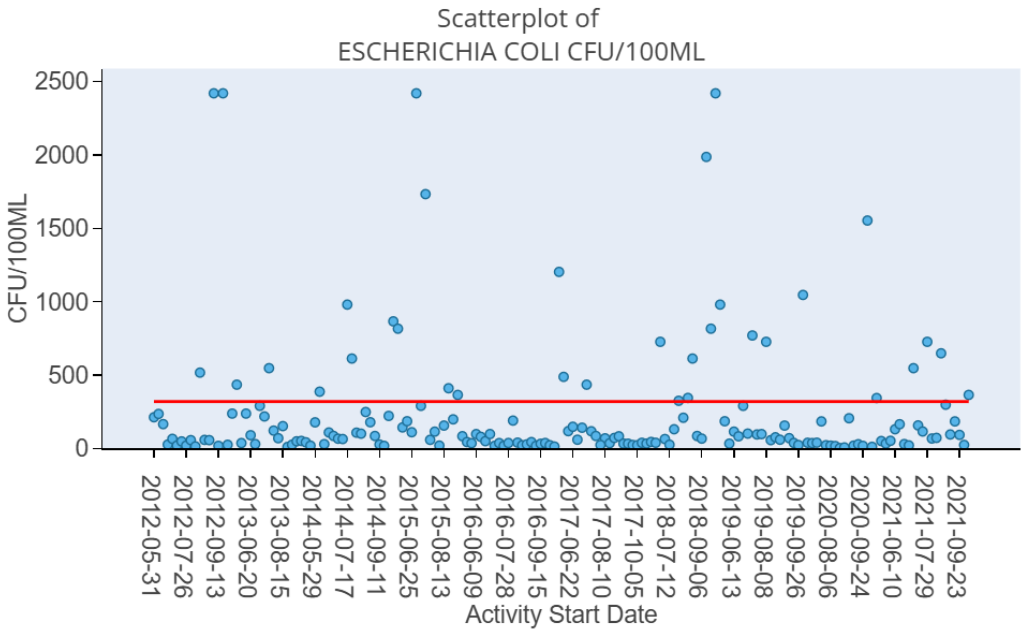
TADA Module 3: Define Water Quality Criteria and Analyze

```
EPATADA::TADA_Scatterplot(ecoli, id_cols = "TADA.ComparableDataIdentifier") |>
  plotly::add_lines(
    y = 320,
    x = c(min(ecoli$ActivityStartDate), max(ecoli$ActivityStartDate)),
    inherit = FALSE,
    showlegend = FALSE,
    line = list(color = "red"),
    hoverinfo = "none"
  )
```

Let's check if any results are above the EPA 304A recommended maximum criteria magnitude (see: 2012 Recreational Water Quality Criteria Fact Sheet).

CRITERIA ELEMENTS	Recommendation 1 Estimated Illness Rate 36/1,000		Recommendation 2 Estimated Illness Rate 32/1,000	
Indicator	GM (cfu/100 mL)	STV (cfu/100 mL)	GM (cfu/100 mL)	STV (cfu/100 mL)
Enterococci (marine & fresh)	35	130	30	110
E. coli (fresh)	126	410	100	320

If interested, you can find other state, tribal, and EPA 304A criteria in EPA's Criteria Search Tool.



TADA Working Group: Identified Requirements/Priorities

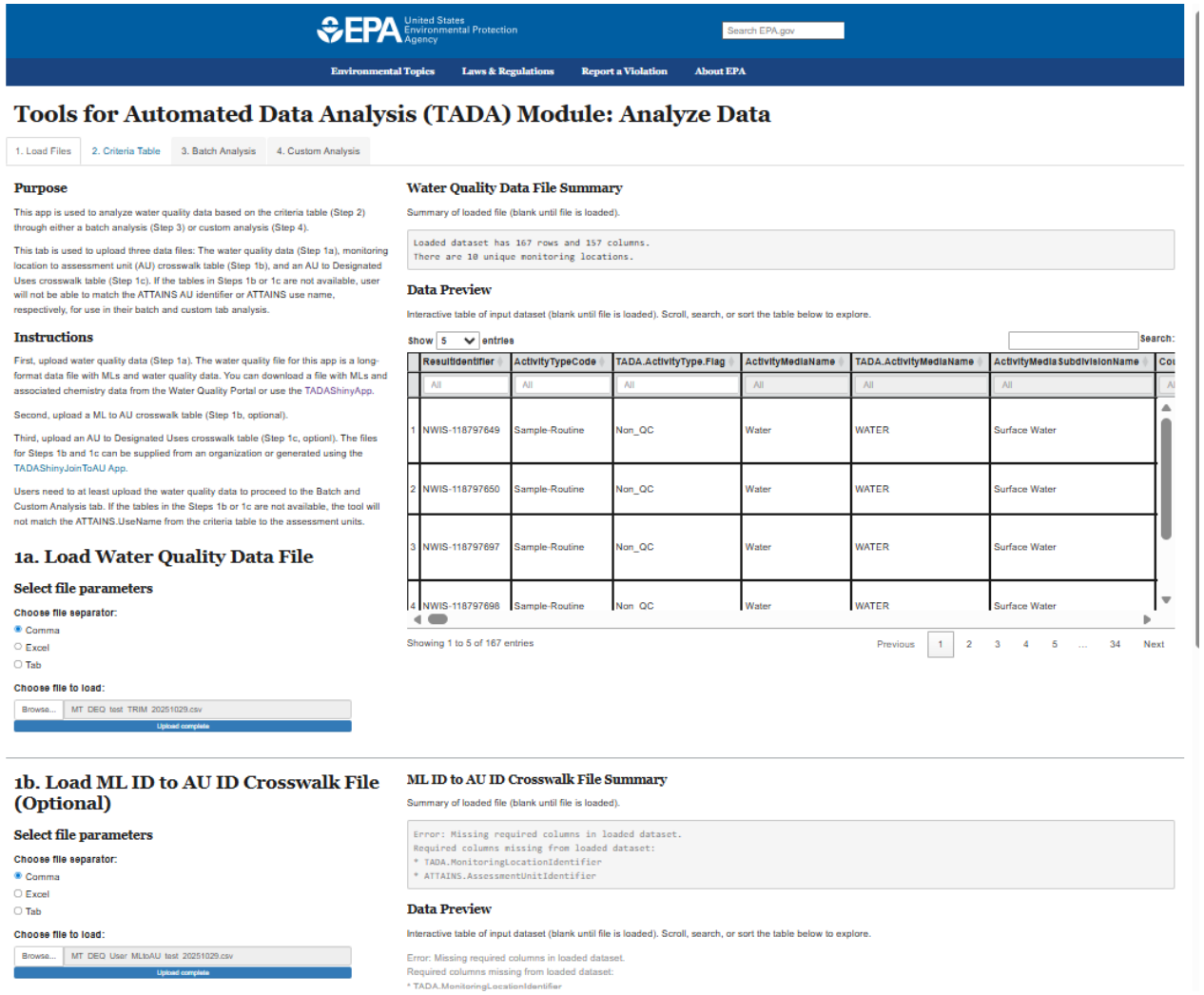
Scope

- Focusing on quantitative (numeric) water data in the WQP to start
- Focusing on frequently assessed parameters
- Common assessment processes and methodologies

Arsenic	Nickel
Boron	Nitrate
Chlorophyll a	Total Nitrogen, mixed forms
Chromium	pH
Chromium(VI)	Total Phosphorus, mixed forms
Cadmium	Depth, Secchi disk depth
Copper	Selenium
Dissolved oxygen (DO)	Silver
Dissolved oxygen saturation	Temperature, water
Escherichia coli	Total suspended solids
Lead	Chromium(III)
Mercury	Zinc

Common Methodologies

- Spatial aggregation – assessment unit and station level assessments
- Characteristic specific assessment start and end dates
- Magnitude, duration (temporal aggregation)
- Criteria context – upper or lower limit, range
- n-day mean, n-day mean maximum or mean minimum, n-hour mean, geometric mean, arithmetic mean, n-day rolling average
- Frequency criteria (e.g., 10% rule, 1-in-3 years rule applied using binomial test or percentile)
- Custom input equations needed to calculate criteria (e.g., for ammonia and certain metals)
- Incorporating depth
- Acute vs chronic
- Seasonality



https://rconnect-public.epa.gov/TADAShinyAnalyze/

EPATADA 0.0.1

Example Module 3 Workflow

TADA Team

2026-03-02

Source: `vignettes/ExampleMod3Workflow.Rmd`

Define Spatial Summary by Monitoring Location (ML)

Let's start off with a ML summary of MTDEQ data frame. By default, we will only display rows for parameters and uses for a ML if it contains data collected for that TADA.CharacteristicName in your WQP data query. If you would like to display all information even for sites with no WQP data, please choose displayNA = TRUE

```
MT.MLSummaryRef.ML <- TADA_MLSummary(
  .data = tada.MT.clean,
  usesRef = MT.usesRef.Final,
  org_id = "MTDEQ",
  displayNA = TRUE,
  excel = FALSE
  # excel = TRUE, overwrite = TRUE
)
```

```
## [1] "TADA_MLSummary: displayNA = TRUE: This MLSummaryRef table
```

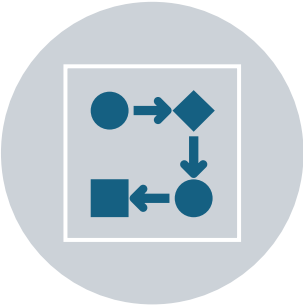
https://usepa.github.io/EPATADA/articles/ExampleMod3Workflow.html

TADA Module 3: Define Water Quality Criteria and Analyze

Two R Package Vignettes

EXAMPLE MODULE 3 WORKFLOW

(DETAILED)



EXAMPLE MODULE 3 WORKFLOW

(STREAMLINED)

R Shiny Apps

- Public web app not yet available (under development, open to early feedback, aiming to make available by the end of 2026).
- GitHub Repository (users can run the app from R Studio)

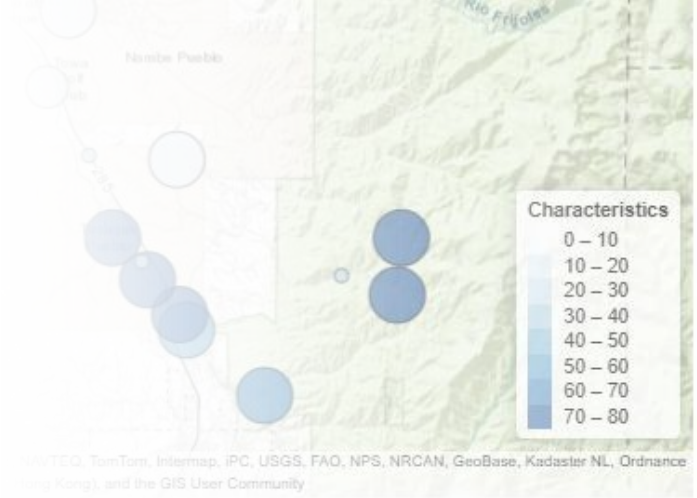
TADA Community Hub

A collaborative hub where TADA users share and maintain information for custom workflows (e.g., state, tribal, territory, EPA, or other entities).

Includes user-contributed criteria and methodologies templates to support reproducible and efficient analyses.

Welcome any states or tribes to reach out to get started on creating a criteria table for your organization.

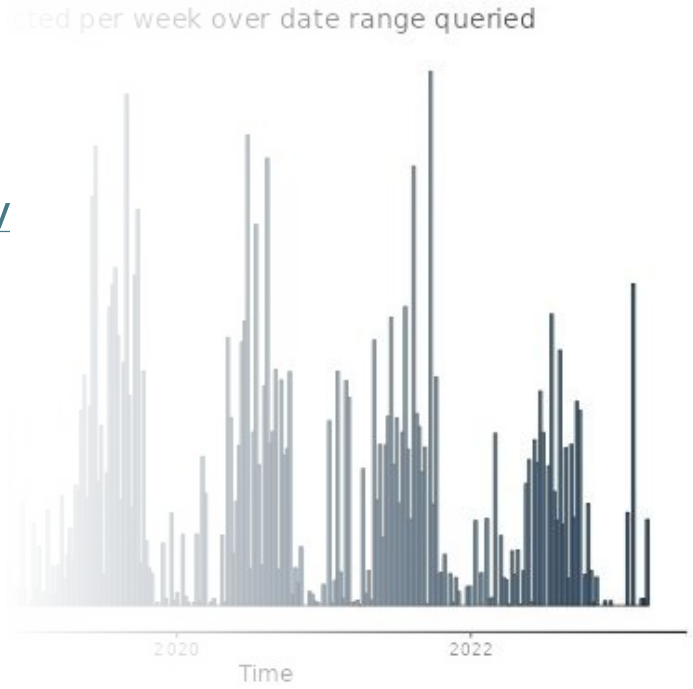
Questions?

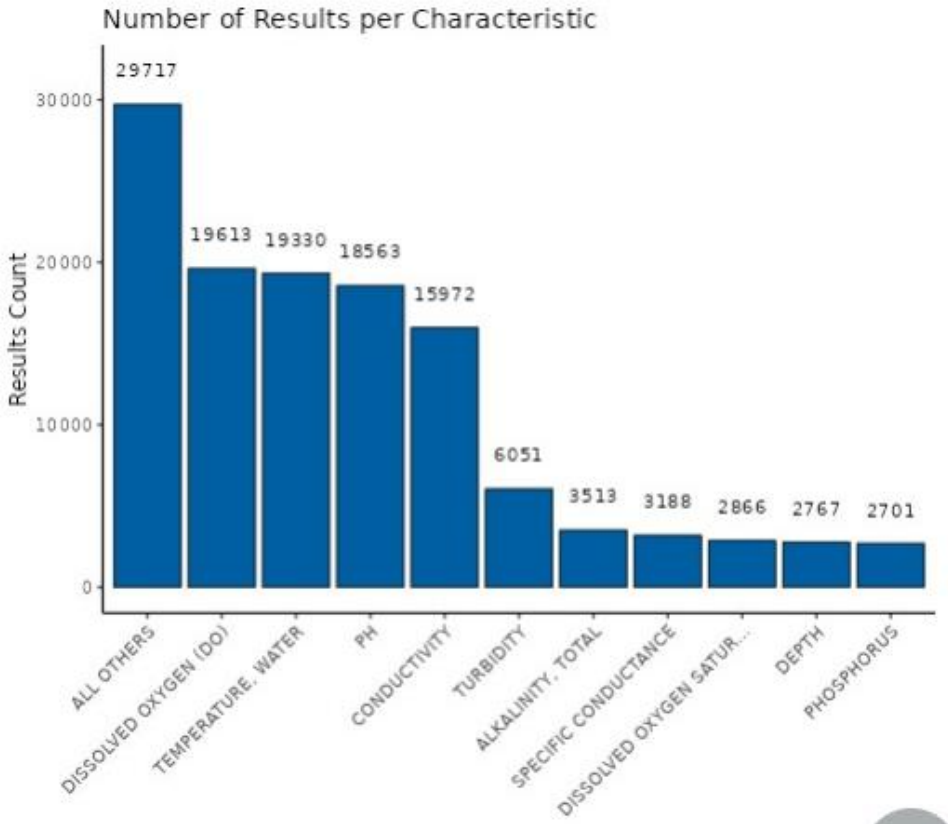


OrganizationFormalName	Result_Count
Chickasaw Nation Environmental Service	4580
Fond du Lac Band of Chippewa (MN)	20176
Pueblo Of Tesuque	6795
Pueblo of Pojoaque	1181
Red Lake DNR	81734
Sac and Fox Nation (Tribal)	9815

Showing 1 to 6 of 6 entries Previous 1 Next

- Contact TADA Team: mywaterway@epa.gov
- Cristina Mullin: Mullin.Cristina@epa.gov





https://www.epa.gov/waterdata/TADA

Up Next: R Workflow Demo Using Penobscot Nation's Publicly Available WQP Data

https://usepa.github.io/EPATADA/articles/PenobscotNationWorkflow.html